

OWATONNA DEGNER, MN, USA

WMO: 726568

Lat: 44.123N

Lon: 93.261W

Elev: 349

StdP: 97.20

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04961

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.5	-29.1	0.3	-24.3	-27.0	0.3	-22.4	12.9	-9.1	11.9	-7.8	2.8	280	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.2	23.4	30.2	22.5	28.7	21.8	25.6	29.6	24.3	28.3	23.3	27.1	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.8	28.1	22.9	18.4	26.8	22.2	17.7	26.3	80.8	29.5	75.2	28.6	71.3	27.0	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.6	9.2	-28.5	34.9	3.1	1.9	-30.8	36.2	-32.6	37.4	-34.3	38.4	-36.6	39.8		
(5)			-28.8	27.4	3.1	1.4	-31.0	28.5	-32.8	29.3	-34.5	30.1	-36.7	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-9.3	-7.2	0.1	7.9	14.7	20.2	22.1	20.7	16.9	9.5	1.7	-5.9		
	DBStd	12.39	7.45	7.25	7.04	5.72	4.99	3.74	3.16	3.12	4.85	5.34	6.30	6.82		
	HDD10.0	2341	597	481	314	104	14	0	0	0	4	74	258	494		
	HDD18.3	4240	856	714	565	316	136	23	5	12	83	278	500	752		
	CDD10.0	1501	0	0	8	41	158	307	376	333	212	59	8	0		
	CDD18.3	359	0	0	1	2	22	80	122	87	41	4	0	0		
	CDH23.3	3184	0	0	3	32	248	697	1124	679	355	46	1	0		
	CDH26.7	927	0	0	0	8	82	217	349	166	97	9	0	0		
(14) Wind	WSAvg	4.1	4.6	4.6	4.6	5.0	4.7	3.9	3.1	2.8	3.4	4.1	4.6	4.4		
(15) Precipitation	PrecAvg	871	23	19	53	75	105	130	113	126	88	62	42	29		
	PrecMax	1044	75	46	97	147	194	218	213	262	285	155	133	80		
	PrecMin	584	5	2	9	19	40	69	34	42	22	13	4	7		
	PrecStd	115	18	12	24	35	45	43	51	50	61	39	32	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	12.0	22.1	27.5	32.5	33.5	33.3	32.3	32.2	27.8	20.2	12.4		
		MCWB	3.8	9.1	15.0	16.8	19.6	23.2	25.5	24.4	21.8	17.6	13.2	10.0		
	2%	DB	3.6	7.3	17.1	22.8	28.7	31.3	32.1	30.0	29.0	23.9	16.4	7.4		
		MCWB	1.9	5.5	12.8	13.8	18.0	21.9	25.1	23.7	21.1	16.0	11.0	5.3		
	5%	DB	2.3	3.9	12.9	20.0	26.2	28.9	30.1	28.0	27.4	21.2	13.0	4.1		
		MCWB	1.1	2.4	9.5	12.5	16.7	20.7	23.7	22.5	19.8	15.0	9.1	2.3		
	10%	DB	1.0	2.4	9.8	17.4	22.9	27.4	28.6	27.3	25.2	17.9	10.9	2.6		
		MCWB	0.0	1.2	7.1	11.0	15.3	19.8	22.9	21.8	18.9	13.2	7.6	1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	8.9	17.2	18.3	22.5	26.0	27.8	26.6	24.4	20.9	15.0	10.5		
		MCDB	6.2	10.8	19.7	23.3	29.1	30.2	31.3	29.8	28.8	24.6	19.2	12.3		
	2%	WB	2.2	5.3	13.5	15.8	20.2	24.1	26.0	25.0	22.9	18.0	12.4	5.3		
		MCDB	3.3	7.1	16.1	21.0	25.7	28.5	30.3	28.6	27.1	20.9	15.4	7.3		
	5%	WB	0.9	2.8	10.1	13.7	18.7	22.6	24.7	23.9	21.6	15.8	9.8	3.0		
		MCDB	2.0	3.9	12.8	18.5	23.1	27.2	28.8	27.2	25.7	20.0	13.0	4.1		
	10%	WB	-0.2	1.2	7.0	11.7	17.2	21.2	23.6	22.8	20.1	13.8	7.6	1.4		
		MCDB	0.6	2.3	9.4	16.2	21.2	25.7	27.5	25.9	24.1	17.9	9.8	2.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.9	8.9	10.8	11.2	10.6	10.8	10.6	11.7	10.7	8.7	7.8		
		MCDBR	8.8	9.1	12.7	15.5	15.0	12.7	11.3	11.4	13.0	14.4	12.8	9.5		
		MCWBR	7.8	7.6	8.8	8.5	6.8	5.8	6.1	6.4	6.4	7.8	8.4	7.8		
		MCDBR	8.3	8.2	11.9	13.4	11.8	10.8	10.3	9.9	10.9	11.4	11.5	8.5		
	5% WB	MCWBR	7.5	7.0	8.8	8.6	6.7	6.4	6.3	6.3	6.5	7.6	8.0	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.289	0.315	0.369	0.396	0.408	0.408	0.402	0.373	0.324	0.291	0.267		
	taud		2.426	2.354	2.409	2.310	2.290	2.297	2.297	2.312	2.364	2.496	2.503	2.448		
	Ebn at Noon		871	906	925	894	875	863	858	851	849	857	836	836		
	Edh at Noon		72	93	102	122	129	128	127	121	106	81	67	64		
(44) All-Sky Solar Radiation	RadAvg		1.76	2.83	3.73	4.59	5.37	6.00	6.34	5.32	4.21	2.72	1.77	1.28		
	RadStd		0.20	0.30	0.37	0.49	0.43	0.48	0.26	0.37	0.30	0.36	0.16	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page